04/02/2026: Classic Score Matching

Reference: Hyvärinen, A. (2005). *Estimation of non-normalized statistical models by score matching.*

CSCI 546: Diffusion Models

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Course Logistics

Final Project Notes

Announcement

I've provided specs about the final project in the syllabus.

Reading Survey

1. Describe score matching. For example, how does it work and when would you use it?

Mini-Lecture: Diffusion

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See previous slide deck

Group Work

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See previous slide deck

Mini-Lecture: Score Matching

A fundamental problem

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Any set of non-negative functions $\{q_{\boldsymbol{\theta}}(\boldsymbol{x})\}_{\boldsymbol{\theta} \in \Theta}$ parameterized by $\boldsymbol{\theta}$ can be used to construct $\{p_{\boldsymbol{\theta}}(\boldsymbol{x})\}_{\boldsymbol{\theta} \in \Theta}$, a family of probability densities over $\boldsymbol{X}$.

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Since probability density functions **must** integrate to 1, we set

$$p_{\boldsymbol{\theta}}(\boldsymbol{x}) = \frac{q_{\boldsymbol{\theta}}(\boldsymbol{x})}{Z_{\boldsymbol{\theta}}},$$

where $Z_{\boldsymbol{\theta}} := \int q_{\boldsymbol{\theta}}(\boldsymbol{x}) d\boldsymbol{x}$ is a normalizing constant.

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The problem. What if we can't compute $Z_{\boldsymbol{\theta}}$?

A solution. We show how to train and sample from these models using score matching, which matches the **score functions** of the model to the data.

Score function

Definition. The **score function** (or *Stein score*) is the gradient of the log likelihood with respect to the data vector $\boldsymbol{x}$:

$$\boldsymbol{s}(\boldsymbol{x}) = \nabla_{\boldsymbol{x}} \ln p_{\boldsymbol{\theta}}(\boldsymbol{x})$$

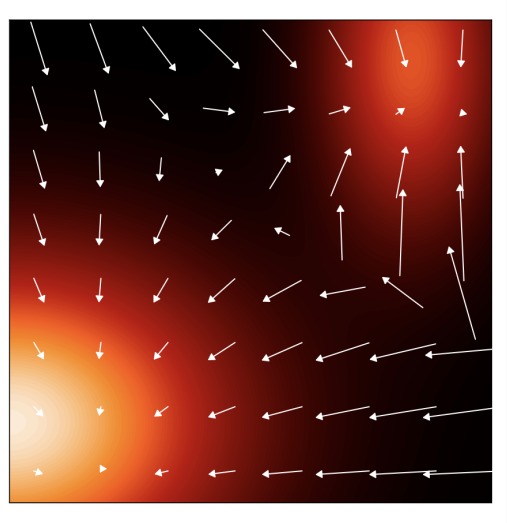

Figure 1: Illustration of score function for a mixture of two-dimensional Gaussians.

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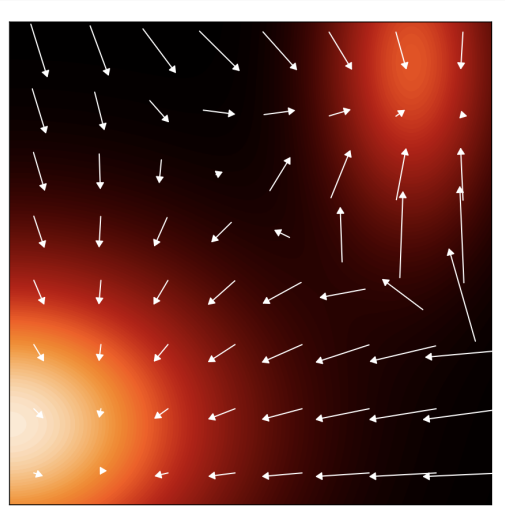

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Poll. Interpretation?

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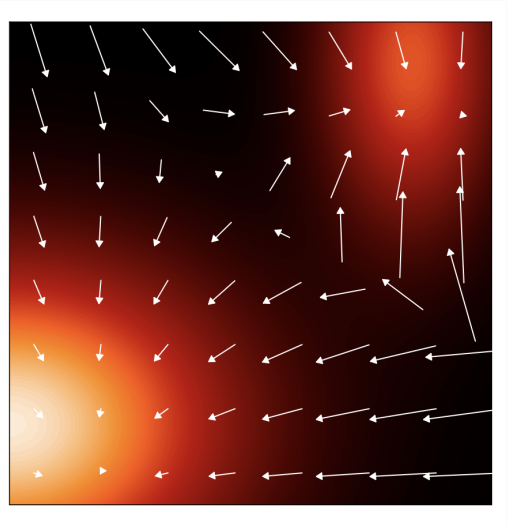

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Poll. Interpretation? The score function is a vector field pointing to the direction of steepest increase in the log likelihood.

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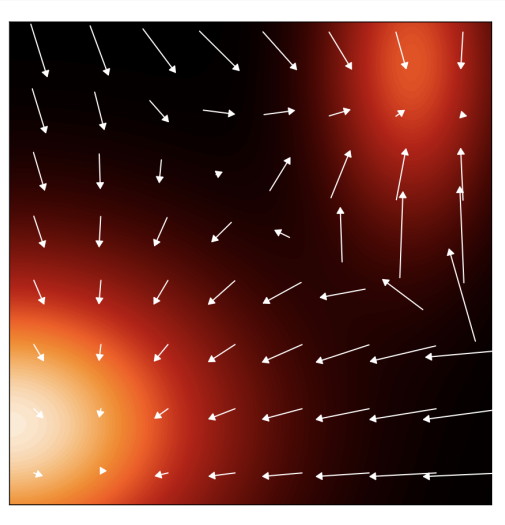

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Poll. Why is this useful?

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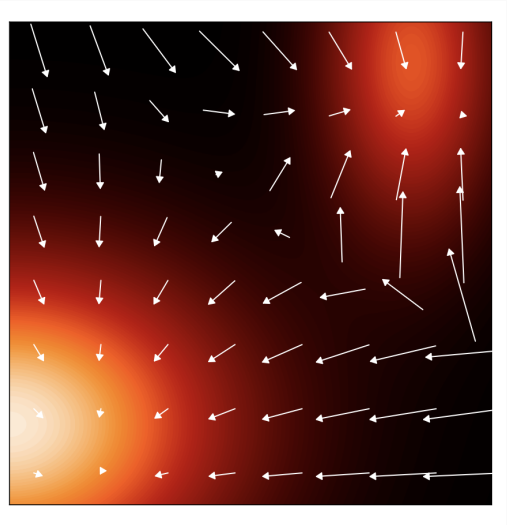


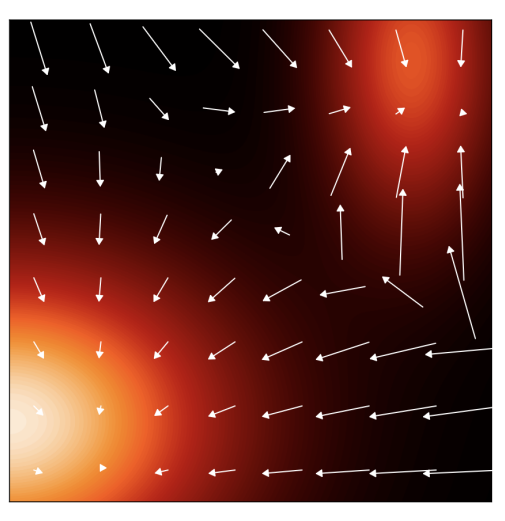
Figure 1: Illustration of score function for a mixture of two-dimensional Gaussians.

Poll. Why is this useful? We don't need the normalizing constant!

$$\nabla_{\boldsymbol{x}} \ln p_{\boldsymbol{\theta}}(\boldsymbol{x}) = \nabla_{\boldsymbol{x}} \ln q_{\boldsymbol{\theta}}(\boldsymbol{x}) - \cancel{\nabla_{\boldsymbol{x}} \ln Z_{\boldsymbol{\theta}}} = \nabla_{\boldsymbol{x}} \ln q_{\boldsymbol{\theta}}(\boldsymbol{x}).$$

Langevin Sampling

Poll. How can we use the score function to obtain a sample $\boldsymbol{x}$ from the probability distribution with density p?



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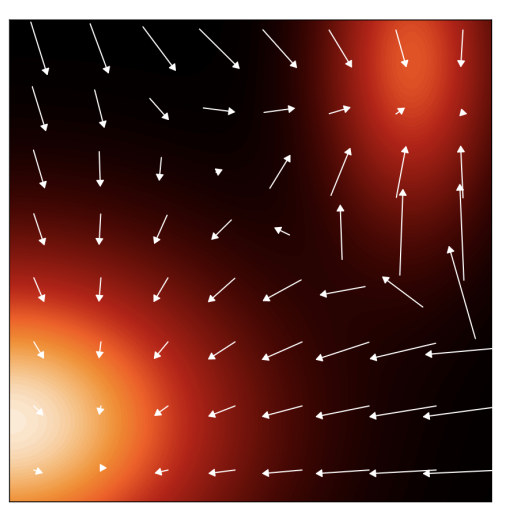

Solution. Start with an initial value $\boldsymbol{x}^{(0)}$, and then iterate the following Markov chain steps:

$$\boldsymbol{x}^{(t)} = \boldsymbol{x}^{(t-1)} + \eta \underbrace{\nabla_{\boldsymbol{x}} \ln p_{\boldsymbol{\theta}}(\boldsymbol{x})}_{\text{score function}} + \sqrt{2\eta}\boldsymbol{\epsilon}^{(t)}, \qquad t = 1, 2, \ldots T.$$

where $\boldsymbol{\epsilon}^{(t)} \sim \mathcal{N}(0, \boldsymbol{I})$ and η is a learning rate.

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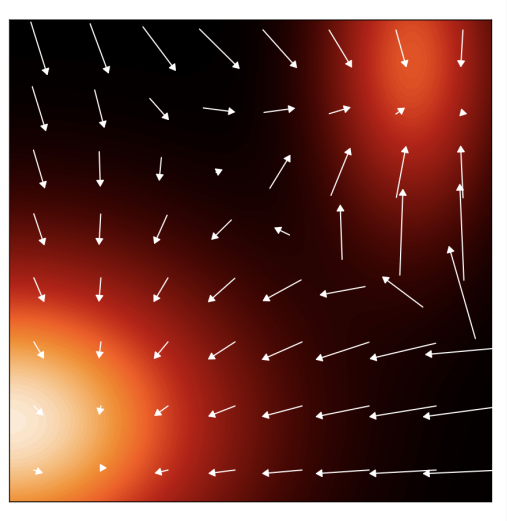

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where $\boldsymbol{\epsilon}^{(t)} \sim \mathcal{N}(0, \boldsymbol{I})$ and η is a learning rate.

Theorem. As $T \to \infty$, $\boldsymbol{x}^{(T)}$ becomes an independent sample from $p_{\boldsymbol{\theta}}(\boldsymbol{x})$.

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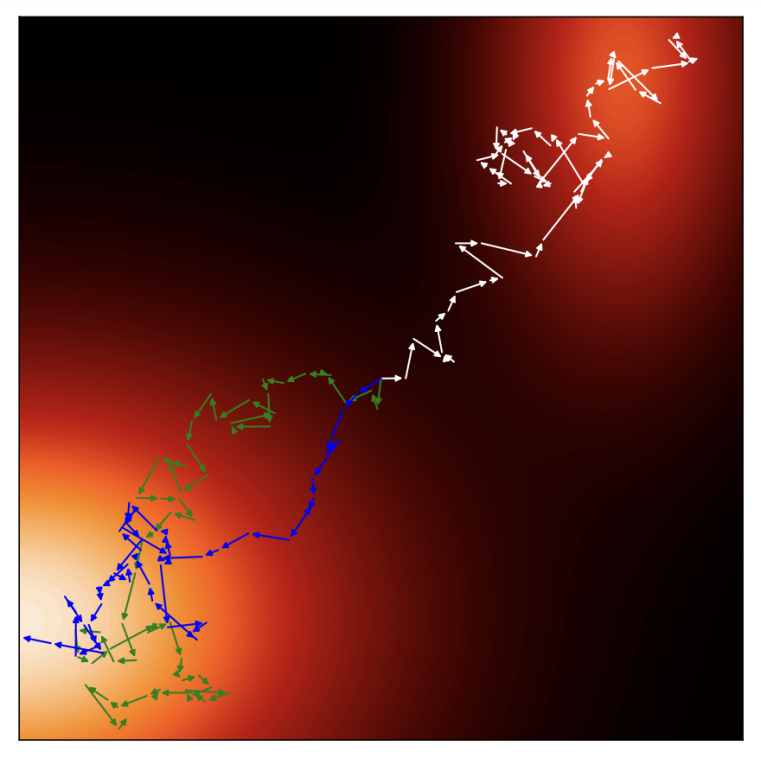

Figure 2: Three trajectories, all starting from the center of the plot.

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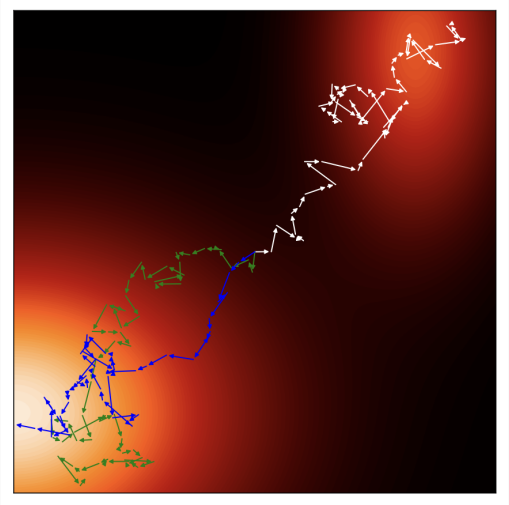

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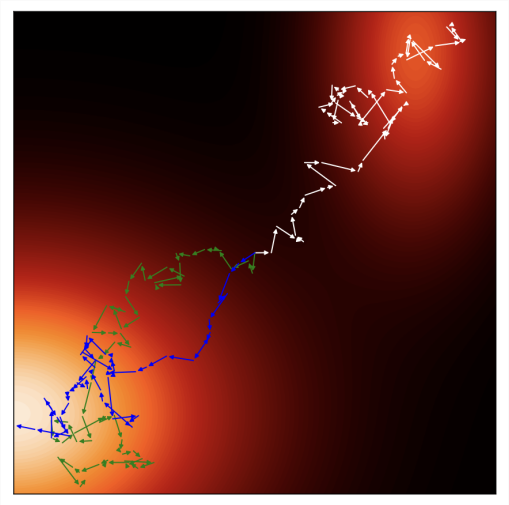

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Solution. We don't need the probability distribution to be normalized!

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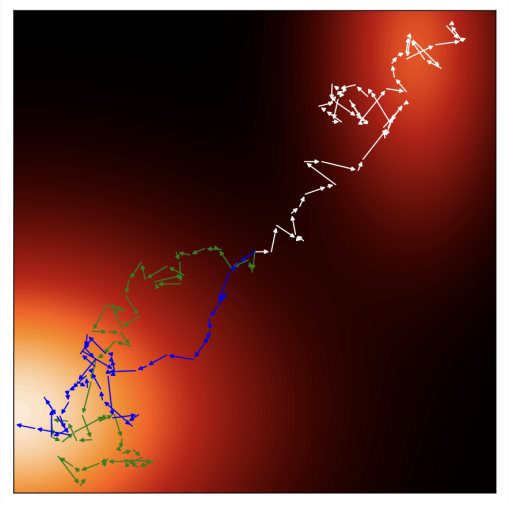

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Problem. But how do we learn $\nabla_{\boldsymbol{x}} \ln p_{\boldsymbol{\theta}}(\boldsymbol{x})$ from data?

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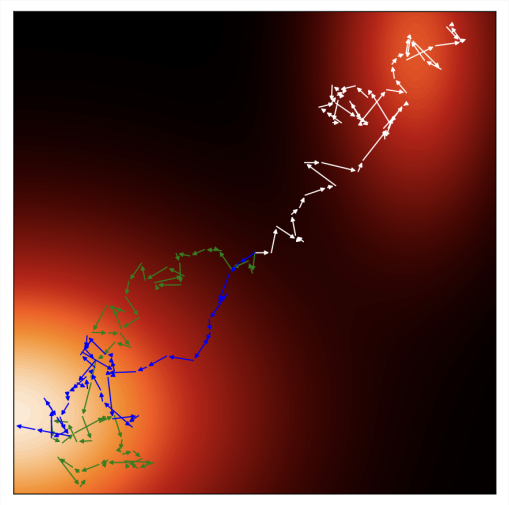

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Solution. `Score matching` to the rescue!

Score Matching

Let

- $p_0(\boldsymbol{x})$ be the density of the true data generating distribution
- $p_{\boldsymbol{\theta}}(\boldsymbol{x})$ be the density of our model.

Score Matching

Then the **score matching** training loss is given by

$$J(\boldsymbol{\theta}) = \int \frac{1}{2} \| \underbrace{\nabla_{\boldsymbol{x}} \ln p_{\boldsymbol{\theta}}(\boldsymbol{x})}_{\text{model score}} - \underbrace{\nabla_{\boldsymbol{x}} \ln p_0(\boldsymbol{x})}_{\text{true score}} \|^2 \, p_0(\boldsymbol{x}) \, d\boldsymbol{x}$$

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Poll. What is the interpretation of this training loss?

Solution. It's the expected squared error between the model score and the true score. (Multiplied by a half).

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Idea. If we want to learn a modeled score function in a flexible manner, we could train a neural network $s(\boldsymbol{x}, \boldsymbol{\theta}) = \nabla_{\boldsymbol{x}} \ln p_{\boldsymbol{\theta}}(\boldsymbol{x})$ within this loss function.

Problem. There's a blocker to doing this. What is it?

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Problem. There's a blocker to doing this. What is it? Even if we have collected a bunch of observations $\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots \boldsymbol{x}_n$, there's no way to know the true score of these observations!

Hyvärinen's Solution

Problem. We want to train a neural network $s(\boldsymbol{x}, \boldsymbol{\theta})$ via the loss function

$$J(\boldsymbol{\theta}) = \int \frac{1}{2} \| \underbrace{s(\boldsymbol{x}, \boldsymbol{\theta})}_{\text{model score}} - \underbrace{\nabla_{\boldsymbol{x}} \ln p_0(\boldsymbol{x})}_{\text{true score}} \|^2 \, p_0(\boldsymbol{x}) \, d\boldsymbol{x},$$

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but we don't know the true score .

Solution. Using integration by parts, we can rewrite this as

$$J(\boldsymbol{\theta}) = \int \left\{ \nabla \cdot s(\boldsymbol{x},\boldsymbol{\theta}) + \frac{1}{2} \| s(\boldsymbol{x},\boldsymbol{\theta}) \|^2 \right\} p_0(\boldsymbol{x}) \, d\boldsymbol{x},$$

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Implication. Who cares? Now the integrand doesn't refer to the true score at all!

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Implementation. But we also don't know $p_0(\boldsymbol{x})$. So how to use in practice? If we have N observations from $p_0(\boldsymbol{x})$, then we can use a Monte Carlo approximation:

$$J(\boldsymbol{\theta}) \approx \frac{1}{N} \sum_{i=1}^{N} \left\{ \nabla \cdot s(\boldsymbol{x}_i,\boldsymbol{\theta}) + \frac{1}{2} \| s(\boldsymbol{x}_i,\boldsymbol{\theta}) \|^2 \right\}$$

Lightning Chat

Lightning Chat